A Two-Layer Decision-Making System for Vehicle Lane Changing Based on Reinforcement Learning

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Abstract

Lane changing is a fundamental problem in autonomous driving. This dissertation proposes a two-layer decision-making and control framework for autonomous vehicles to merge into the target lane. The proposed method decomposes the lane changing problem into two coupled but functionally distinct layers: the upper-level module determines which local gap is preferable, while the lower-level module converts the selected gap into continuous target points and actual vehicle control inputs. The upper-level module uses only the three target-lane vehicles closest to the ego vehicle, forms two candidate gaps with these three vehicles, evaluates the relative feasibility of the two gaps through a smooth confidence function, and updates the upper-level opinion state using the opinion dynamics equation with reinforcement terms. The lower-level module further calculates the objective bias based on the size and rate of change of the selected gap, uses a SAC policy to determine the attention intensity, updates the lower-level opinion state, and maps this opinion to the target control points within the gap. Finally, the front-axle tracking controller generates vehicle acceleration and angular velocity changes.

Simulation experiments are implemented in Python to evaluate the proposed control method. The experimental results show that in a single-gap environment, the reinforcement learning method of SAC learns an effective strategy and maintains convergence, and the vehicle can successfully complete the task; in a multi-gap environment, the policy learned in the simple environment transfers to the more complex setting, and the effectiveness of the two-layer decision-making system is verified.

1 Introduction

1.1 Background and Motivation

1.1.1 Decision-making strategies for merge and lane-change interaction

Lane-changing decisions are representative problems in autonomous driving. A lane change alters the available gaps, road right-of-way expectations, and future responses of surrounding vehicles. Recent research has therefore increasingly treated lane changing as an interactive perception decision task. Game theory models are widely used because they can explicitly describe the strategic relationship between the ego vehicle and surrounding vehicles. Lane changing intentions, collision probabilities, and dynamic risks can be incorporated into the payoff function to determine whether a maneuver is beneficial or dangerous [1], [2]. Physics-inspired models such as molecular interaction potentials further express the attraction and repulsion between vehicles as continuous interaction forces [3]. Recent studies also consider incomplete information and social driving preferences, as in lane changing conflicts involving human-driven vehicles, surrounding vehicles may exhibit cooperative, aggressive, or uncertain behaviors [4].

Another important direction is learning-based interactive prediction. Graph neural networks and attention mechanisms represent vehicles as nodes and the influence between vehicles as edges or attention weights. This is highly suitable for lane changing scenarios, as the number and importance of surrounding vehicles will dynamically change. Existing research uses topological graphs combined with driving behavior and traffic context to predict the lane changing intentions of drivers [5], and some studies use plan-aware graph attention networks to predict the responses of surrounding vehicles to the candidate intentions of the ego vehicle, and then the model predicts the control trajectory [6]. These methods illustrate that interaction modeling should not only predict where other vehicles will go, but also estimate how they will respond to the ego vehicle's decision. Cooperative control methods provide a third route. In connected autonomous driving environments, surrounding vehicles can actively create gaps, coordinate lane changing sequences, or maintain fleet stability [7], [8]. Overall, recent literature provides motivation for the hierarchical framework: this framework requires the combination of strategy interaction modeling, behavior prediction, and safe execution.

1.1.2 Reinforcement learning for adaptive decision dynamics

The lane-changing decision-making process exhibits sequentiality, uncertainty, and feedback-driven characteristics. Vehicles need to determine when to wait, when to change lanes, and how to adjust their speed. Traditional fixed-parameter controllers may perform well in static scenarios, but they often lack flexibility when the traffic environment changes or when the system needs to switch between cooperative consensus and competitive divergence. Deep reinforcement learning provides a mechanism for learning decision strategies through repeated interactions. Existing studies have integrated safety rules, future risk assessment, reward shaping, and robust observation modeling into DRL lane-changing strategies [9], [10], [11]. These studies show that RL should not be used as an unconstrained black box, but should be guided by interpretable state variables, safety-aware reward terms, and robust mechanisms.

Meanwhile, MARL allows each connected or autonomous vehicle to learn from the behavior of neighboring vehicles while considering system-level outcomes such as traffic efficiency, comfort, and safety [12]. Right-of-way coordination and Mix Q-learning further illustrate that the lane-changing task can balance individual benefits with group benefits [13], [14]. For this dissertation, these studies support a key idea: RL is not just a direct output of lane-changing actions, but can dynamically adjust the control parameters of the nonlinear decision dynamics model. This enables adaptive changes according to environmental stimuli while still maintaining connection with low-level safety control.

1.1.3 Evaluation dimensions: from efficiency to safety, comfort, and adaptability

The evaluation of lane-changing decisions has shifted from a single efficiency metric to a multi-dimensional assessment. Average speed, travel time, traffic volume, and lane-changing success rate remain important, but they are not sufficient to support safety-critical autonomous driving systems. A rapid lane change can still be unsafe, uncomfortable, or disrupt the traffic flow. Therefore, recent research has begun to evaluate collision risk, minimum distance, braking feasibility, safety violation rate, acceleration, deceleration, traffic flow stability, and robustness under perception uncertainty [7], [8], [10], [15]. This project adopts these evaluation perspectives, which emphasize convergence speed, safety violation rate, and task efficiency while also evaluating safety, comfort, and stability.

1.2 Aims and Objectives

The aim of this project is to develop, integrate, and evaluate a hierarchical decision-making and control framework for interactive autonomous driving scenarios. This framework will use reinforcement learning to adjust the key parameters of the nonlinear decision dynamics/control model, enabling the agent to quickly make decisions in the simulation environment while meeting safety constraints.

This project has five specific objectives. First, it establishes a nonlinear opinion dynamics model at the policy layer. Second, it trains the agent using RL methods to dynamically optimize the model parameters. Third, it develops a safety perception and execution layer, using control barrier functions or equivalent constraints to convert the high-level policy into collision-free trajectories. Fourth, it implements the controller in the Python simulation framework. Fifth, it compares with the baseline and evaluate the decision-control effectiveness.

2 Methods

The controller adopts a two-layer structure. The upper-layer controller completes the discrete control problem of gap selection through the evaluation of different gaps and the update of opinion dynamics; the lower-layer controller, solves the continuous control problem of actual vehicles using the actual size and change rate of the gap, as well as the attention parameters learned by reinforcement learning.

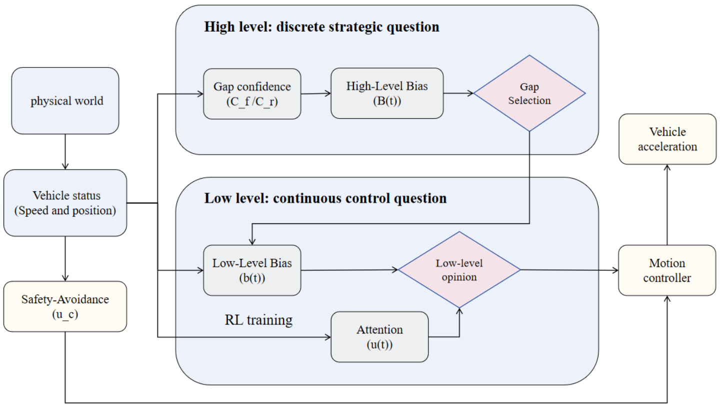

Fig. 1. Control system structure block diagram

2.1 High-Level Control System

2.1.1 Opinion Dynamics and Self-Updating Attention

The same opinion-dynamics template [16] is used at both levels. As shown in the following formula, for the general opinion variable $o(t)$, its dynamics are defined below, where the damping coefficient $d > 0$ pulls the opinion back to the neutral position to prevent it from drifting infinitely; the attention intensity $u(t) \geq 0$ determines the degree to which the existing opinion is self-reinforced; the sensitivity parameter $\alpha > 0$ determines the speed at which the nonlinear term enters the saturation zone; the external bias $b(t)$ injects the environmental evaluation into the system. If $b(t)$ is close to zero, the damping term will cause the opinion to decay; if there is a small but persistent bias, the self-reinforcement term will gradually amplify the opinion until a stable decision is formed.

$$\dot{o}(t) = -d\,o(t) + u(t)\tanh(\alpha o(t)) + b(t). \tag{1}$$

The dynamics of the high-level decision opinion can be expressed in the following form, where $y(t)$ represents the opinion of the high-level decision-making.

$$\dot{y}(t) = -d_y y(t) + u_h(t)\,\mathrm{tanh}(\alpha_y y(t)) + B(t). \tag{2}$$

In high-level decision-making, the external environment bias is determined by the gap confidence, while attention is updated by the following self-updating law. This Hill-type function depends on y^2, so the sign of the opinion is treated symmetrically. When $y(t)$ approaches zero, attention decays and the vehicle remains cautious; when $|y(t)|$ increases, S_h rises, thereby increasing $u_h(t)$, and a larger $u_h(t)$ further strengthens the self-reinforcing term in the opinion equation. This feedback loop enables small but continuous preferences to gradually transform into stable gap selection.

$$\begin{aligned} \dot{u}_h(t) &= \frac{-u_h(t) + S_h(y(t)^2)}{\tau_h}, \\ S_h(y^2) &= U_{\mathrm{max}} \frac{(y^2)^n}{K_h^n + (y^2)^n}. \end{aligned} \tag{3}$$

2.1.2 High-Level Bias from Gap Confidence

At each time step, the high-level module selects the three target-lane vehicles closest to the ego front axle in the longitudinal coordinate. Let these vehicles be ordered from front to rear as (i_1, i_2, i_3). The forward candidate gap is the space between i_1 and i_2, and the rear candidate gap is the space between i_2 and i_3. For any candidate gap formed by a front vehicle F and a rear vehicle R, the gap center and gap velocity are

$$x_g(t) = \frac{p_F^x(t) + p_R^x(t)}{2}, \quad v_g(t) = \frac{v_F^x(t) + v_R^x(t)}{2}. \tag{4}$$

The ego-to-gap alignment is defined by the longitudinal position error $d_g(t) = x_g(t) - p_e^x(t)$ and the relative speed error $\Delta v_g(t) = v_g(t) - v_e^x(t)$.

The confidence of a gap is then evaluated using a Gaussian radial-basis function:

$$C_g(t) = \mathrm{exp}\left(-\frac{d_g(t)^2}{2\sigma_d^2} - \frac{\Delta v_g(t)^2}{2\sigma_v^2}\right). \tag{5}$$

When the gap center is close to the ego vehicle and the relative speed is low, the confidence level is higher; when the gap is spatially far apart or the speed matching is poor, the confidence level smoothly decreases. The forward and rear confidences are converted into a signed directional bias $B(t) = C_f(t) - C_r(t)$.

The difference is used because they retain both the direction and intensity of the preference. If $B(t) > 0$, it indicates that the front gap is more consistent with the current ego state; if $B(t) < 0$, it indicates that the rear gap is more in line; if $B(t) \approx 0$, it means the evidence is still ambiguous. Even if the absolute values of the two confidence levels are both small, the difference between them can still represent a weak but meaningful relative preference. Through the opinion dynamics equation, this weak preference will only accumulate and gradually become clearer when it persists. This keeps the comparison process smooth and makes the decision changes more stable.

2.1.3 High-Level Opinion Update and Decision Mapping

The high-level opinion $y(t)$ stores the accumulated directional belief. After computing $B(t)$ and updating $u_h(t)$, the system integrates the high-level opinion using an explicit time step Δt, and updates the attention:

$$
\begin{aligned}
y_{k+1} &= y_k + \Delta t \left[-d_y y_k + u_{h,k} \tanh(\alpha_y y_k) + B_k \right], \\
u_{h,k+1} &= u_{h,k} + \Delta t \frac{-u_{h,k} + S_h(y_k^2)}{\tau_h}.
\end{aligned} \tag{6}
$$

The high-level decision mapping is threshold-based: the system selects the forward gap when $y_k > \theta_y$, the rear gap when $y_k < -\theta_y$, and waits when $|y_k| \le \theta_y$.

The waiting interval prevents frequent changes in the selected gap when C_f and C_r are close, allowing vehicles to leave the neutral zone and make a choice only after sufficient evidence has accumulated. Therefore, this mechanism is smoother than the direct instantaneous comparison. Even if the group selection of the vehicle closest to the ego vehicle changes, the decision will not suddenly change due to a jump in confidence, which is more consistent with the actual driving behavior logic.

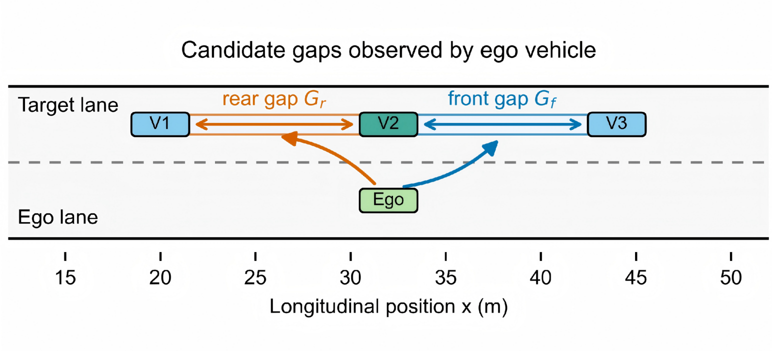

Fig. 2. layered opinion decision schematic

2.2 Low-Level Control System

After the high-level module selects a candidate gap, the low-level module decides how the ego vehicle should enter that gap. The low-level module first evaluates the physical feasibility of the selected gap, then determines the attention intensity, updates the low-level opinion $z(t)$, and finally converts $z(t)$ into continuous target points. This design makes the merging action gradual: as the low-level opinion strengthens, the target points gradually move from the original lane reference position to within the selected gap.

2.2.1 Low-Level Bias

For a selected gap formed by a front vehicle F and a rear vehicle R, the physical gap length and gap-rate term is

$$\begin{aligned} g(t) &= p_F^x(t) - p_R^x(t), \\ \dot{g}(t) &= v_F^x(t) - v_R^x(t), \\ b(t) &= k_g\left[g(t) - g_{\text{safe}}\right] + k_v \dot{g}(t). \end{aligned} \tag{7}$$

The low-level bias evaluates whether the gap is sufficiently large and whether it is opening or closing:

A larger gap makes $b(t)$ more positive. An opening gap, where $\dot{g}(t) > 0$, also increases $b(t)$. A small or closing gap decreases $b(t)$, delaying the growth of the low-level merge intention. This formula is intentionally interpretable: $g(t) - g_{\text{safe}}$ measures spatial feasibility, while $\dot{g}(t)$ measures whether the situation is improving or worsening.

2.2.2 The attention obtained through reinforcement learning

The low-level attention $u(t)$ regulates the intensity of the self-reinforcement of the low-level opinions. However, the optimal timing of attention is difficult to design manually because the system simultaneously contains nonlinear opinion dynamics, moving vehicles, safety avoidance, input saturation, and delay consequences. If attention is increased too early, the ego vehicle may overcommit when the gap is not yet safe; if attention is increased too late, the vehicle may miss the merging opportunity and behave too conservatively. Therefore, reinforcement learning is used to learn the temporal sequence of attention from the reward feedback. At the same time, the learned strategy does not replace the model controller, but only outputs the attention variables in the opinion dynamics. This action space is low-dimensional and interpretable.

2.2.3 SAC Reinforcement Learning Design

Soft Actor-Critic is an off-policy actor-critic algorithm suitable for continuous action spaces. It learns the random policy $\pi_\phi(a|s)$, two soft Q functions, and the entropy temperature parameter. Its optimization objective is to balance between the expected return and the policy entropy:

The entropy term promotes exploration, which is crucial for the lane-changing task because a useful policy not only needs to learn to stay safe but also needs to learn when to actively make a lane change.

$$J(\pi) = \sum_t \mathbb{E}_{(s_t,a_t)\sim\rho_\pi} \left[r(s_t, a_t) + \alpha_{\text{SAC}} \mathcal{H}\left(\pi(\cdot|s_t)\right)\right]. \tag{8}$$

The low-level reinforcement-learning state describes the relative geometry and velocity between the ego vehicle and the front and rear vehicles of the selected gap. The state is defined as follows. Here F and R denote the front and rear vehicles of the selected gap.

$$\begin{aligned} s_t &= \left[\Delta x_F \quad \Delta y_F \quad \Delta v_F^x \quad \Delta v_F^y \quad \Delta x_R \quad \Delta y_R \quad \Delta v_R^x \quad \Delta v_R^y\right]^\top, \\ a_t &= u(t), \qquad u(t) \in [u_{\min}, u_{\max}]. \end{aligned} \tag{9}$$

The policy output is the low-level attention:

The reward function is designed to encourage timely merging, discourage hesitation, penalize oscillatory motion, and strongly penalize collision. A representative per-step reward is

$$\begin{aligned} r_t =& w_p \,\Delta P_t + w_o \,O_t \max(\Delta P_t, 0) - w_h \,O_t(1-P_t) - w_s(1-P_t) \\ &- w_a \|a_t - a_{t-1}\|_2^2 - w_f \,\mathbb{I}[v_y(t)v_y(t-1) < 0] \\ &- w_c \left[\max\left(0, \frac{d_{\text{safe}} - d_{\min}(t)}{d_{\text{safe}}}\right)\right]^2 - 1000\,\mathbb{I}_{\text{collision}} + R_{\text{success}}(t)\,\mathbb{I}_{\text{success}}. \end{aligned} \tag{10}$$

Lane-changing progress is defined as the normalized ratio:

$$P_t = \text{clip}\left(\frac{y_e(t) - y_O}{y_T - y_O}, 0, 1\right), \qquad \Delta P_t = P_t - P_{t-1}. \tag{11}$$

The opportunity term O_t represents whether the selected gap is currently suitable for merging. The progress terms reward moving toward the target lane, especially when a useful opportunity exists. The hesitation terms penalize remaining far from the target lane, so the policy cannot receive a high score by simply waiting. The action-smoothness and direction-flip terms reduce repeated acceleration or lateral-direction reversals. The safety term is continuous before collision, while the collision term is a large terminal penalty. The success bonus is implemented as a time-decreasing term $R_{\text{success}}(t) = 100 - 2t$.

2.2.4 Low-Level Opinion and Target Point

The low-level opinion $z(t)$ represents the degree of merge commitment for the selected gap. Its continuous dynamics and discrete update are

$$\begin{aligned} \dot{z}(t) &= -d_z z(t) + u(t)\tanh(\alpha_z z(t)) + b(t), \\ z_{k+1} &= z_k + \Delta t\left[-d_z z_k + u_k \tanh(\alpha_z z_k) + b_k\right]. \end{aligned} \tag{12}$$

Table 1. Reward terms and their control meanings

Reward term	Meaning
$w_p \Delta P_t$	Rewards direct lane-change progress
$w_o O_t \max(\Delta P_t, 0)$	Gives extra reward for moving during a valid opportunity
$-w_h O_t (1 - P_t)$	Penalizes hesitation when a gap is available
$-w_s (1 - P_t)$	Applies a general time / inaction penalty
$-w_a \|a_t - a_{t-1}\|_2^2$	Penalizes abrupt action changes
$-w_f \mathbb{I}[v_y(t) v_y(t-1) < 0]$	Penalizes lateral direction flips
Safety distance penalty	Penalizes near-collision continuously
Collision penalty	Strong terminal penalty
Success bonus	Rewards early safe completion

A low or negative $z(t)$ keeps the desired target point close to the original lane or a cautious reference. A high positive $z(t)$ moves the target point toward the selected gap in the target lane. To keep this transition bounded, $z(t)$ is mapped through the smooth gate $\lambda_z(t) = \text{clip}((z(t) - z_{\text{min}})/(z_{\text{max}} - z_{\text{min}}), 0, 1)$.

2.3 Actual Control Input

The final control layer converts the desired target point into actual vehicle input. This layer consists of three parts: a safety obstacle avoidance item that pushes the ego vehicle away from the nearby target lane vehicles, a tracking error pointing to the desired target point, and a reverse solution of the bicycle model that maps the desired front axle acceleration to longitudinal acceleration and angular rate of change. Through explicit safety constraints and vehicle kinematics expressions, the entire system becomes safer and more interpretable.

2.3.1 Safety-Avoidance Term

For each surrounding target-lane vehicle j, define the relative vector from that vehicle to the ego front axle as $\mathbf{r}_j(t) = \mathbf{p}_e(t) - \mathbf{p}_j(t)$, with distance $d_j(t) = \|\mathbf{r}_j(t)\|_2$.

The safety-avoidance acceleration is constructed as a repulsive field:

$$\mathbf{u}_c(t) = \sum_j k_c \max\left(0, \frac{d_{\text{safe}} - d_j(t)}{d_{\text{safe}}}\right)^2 \frac{\mathbf{r}_j(t)}{d_j(t)}. \tag{13}$$

When all vehicles are farther than d_{safe}, this term is zero. When a vehicle enters the safety region, the repulsive magnitude grows quadratically as distance decreases. This term does not replace collision checking; instead, it provides a continuous pre-collision correction that can steer the ego vehicle away before a hard collision boundary is reached.

2.3.2 Target Point and Tracking Error

The selected-gap target point is usually placed near the longitudinal center of the gap and at the target-lane center. The original-lane reference point can be placed ahead of the ego vehicle along the original lane:

$$\mathbf{p}_G^\star(t) = \begin{bmatrix} x_g(t) \\ y_T \end{bmatrix}, \quad \mathbf{p}_O^\star(t) = \begin{bmatrix} p_e^x(t) + \ell_{\text{look}} \\ y_O \end{bmatrix}. \tag{14}$$

The opinion-weighted target point and tracking error are:

$$\begin{aligned} \mathbf{p}^\star(t) &= [1 - \lambda_z(t)]\, \mathbf{p}_O^\star(t) + \lambda_z(t)\mathbf{p}_G^\star(t), \\ \mathbf{e}_z(t) &= \mathbf{p}^\star(t) - \mathbf{p}_e(t). \end{aligned} \tag{15}$$

This construction gives the low-level opinion a direct geometric meaning. When $\lambda_z(t)$ is close to zero, the controller behaves like a lane-keeping controller. When $\lambda_z(t)$ approaches one, the controller behaves like a gap-entry controller.

2.3.3 Final Physical Input Design

The desired front-axle acceleration combines proportional position tracking, velocity damping, and safety avoidance as $\mathbf{u}_{\text{total}}(t) = k_p \mathbf{e}_z(t) - k_v \mathbf{v}_e^f(t) + \mathbf{u}_c(t)$.

This vector is then converted into physical inputs. Let the front-axle velocity direction and its normal direction be

$$\begin{aligned} \mathbf{t}_e(t) &= \frac{\mathbf{v}_e^f(t)}{\|\mathbf{v}_e^f(t)\|_2 + \epsilon}, \qquad \mathbf{n}_e(t) = \begin{bmatrix} -t_e^y(t) \\ t_e^x(t) \end{bmatrix}, \\ a(t) &= \text{clip}\left(\mathbf{u}_{\text{total}}(t)^\top \mathbf{t}_e(t), a_{\min}, a_{\max}\right), \\ \omega(t) &= \text{clip}\left(k_\omega\, \mathbf{u}_{\text{total}}(t)^\top \mathbf{n}_e(t), \omega_{\min}, \omega_{\max}\right). \end{aligned} \tag{16}$$

This PID-like structure provides a clear and stable mapping from the opinion-driven target point to the executable vehicle input, while avoiding unrealistic acceleration and steering changes through input clipping.

3 Experimental Design

The experiments are organized from simple to complex. The single gap environment isolated the low-level control problem: testing the learning effect of RL when the target gap is fixed, and whether the controller can learn when to increase attention and commit to merging. The multi-gap environment introduced high-level decisions: when multiple dynamic gaps exist simultaneously, whether the system can select the appropriate local gap to avoid excessive switching and safely execute merging. The experimental parameters are listed in the appendix.

3.1 Single-Gap Merging Experiment

The single-gap environment contains one front vehicle, one rear vehicle, and one ego vehicle. The front and rear vehicles travel in the target lane, and their longitudinal separation defines the only available merging gap. The ego vehicle starts from the original lane and attempts to merge into this gap. The lane width is $W = 4.0\,\mathrm{m}$. The initial target-vehicle states are $x_F(0) = 30.0\,\mathrm{m}$, $x_R(0) = 15.0\,\mathrm{m}$, and $v_F(0) = v_R(0) = 15.0\,\mathrm{m/s}$.

The ego vehicle has the same initial longitudinal speed but a randomized longitudinal initial position, $x_e(0) = 20.0 + \xi$, $\xi \sim \mathcal{U}(-5.0, 5.0)$, $y_e(0) = 2.0\,\mathrm{m}$, and $v_e(0) = 15.0\,\mathrm{m/s}$. This randomization makes the training and evaluation less dependent on one special initial alignment. It forces the policy to learn an attention schedule that works when the ego vehicle starts slightly ahead of or behind the gap center.

The rear vehicle is deliberately designed to create a staged merging opportunity. Before $t_y = 20.0\,\mathrm{s}$, the rear vehicle follows a sinusoidal acceleration law.

$$
\begin{aligned}
\omega_s &= \frac{2\pi}{P_s}, \qquad P_s = 6.0\,\mathrm{s}, \qquad A_s = 4.0\,\mathrm{m/s}, \\
a_R(t) &= A_s \omega_s \cos(\omega_s t), \\
v_R(t) &= v_R(0) + A_s \sin(\omega_s t), \\
x_R(t) &= x_R(0) + v_R(0)t + \frac{A_s}{\omega_s}\left[1 - \cos(\omega_s t)\right], \\
g(t) &= x_F(t) - x_R(t) = 15.0 - \frac{A_s}{\omega_s}\left[1 - \cos(\omega_s t)\right], \quad 0 \le t \le 20.0.
\end{aligned} \tag{17}
$$

After $20.0\,\mathrm{s}$, the rear vehicle starts yielding by tracking a desired front-rear gap $g_{\text{yield}} = 20.0\,\mathrm{m}$. The rear-vehicle acceleration becomes:

$$
\begin{aligned}
a_R(t) &= \mathrm{clip}\left(0.35\left[g(t) - 20.0\right] - 1.1\left[v_R(t) - v_F(t)\right], -5.0, 2.0\right), \quad t > 20.0, \\
\dot{g}(t) &= v_F(t) - v_R(t), \qquad \ddot{g}(t) = -a_R(t), \qquad t > 20.0.
\end{aligned} \tag{18}
$$

This piecewise construction has a clear purpose. Before $20.0\,\mathrm{s}$, the gap is not intentionally opened for the ego vehicle, so the policy should avoid premature aggressive merging. After $20.0\,\mathrm{s}$, the rear vehicle creates a larger gap and the correct behavior is to increase attention $u(t)$, allow the opinion $z(t)$ to grow, and move the target point toward the target lane.

The SAC agent uses a Gaussian policy with two hidden layers of width 256, two Q networks, target Q networks, replay-buffer learning, and entropy regularization. The main training parameters are

200 episodes, replay-buffer size 2.0×10^5, batch size 256, 1000 initial random steps, discount factor $\gamma = 0.99$, target-update rate $\tau = 0.005$, and learning rates 3×10^{-4} for the policy, Q networks, and entropy temperature. The subsequent value comparison is also carried out using the same ”reward” as defined in the previous text. The same reward definition is used during value comparison so that the learned and hand-designed policies are judged by an identical metric. The baseline comparison evaluates the trained SAC attention against the original hand-designed RBF attention:

$$u_{\mathsf{RBF}}(t) = u_{\mathsf{base}} + u_{\mathsf{amp}} \exp\left(-\frac{d_g(t)^2}{2\sigma_d^2} - \frac{\Delta v_g(t)^2}{2\sigma_v^2}\right). \tag{19}$$

The evaluation repeats the simulation 100 times with shared random ego initial positions. For each random seed, both policies face the same initial $x_e(0)$ and the same target-vehicle trajectory. The comparison reports each episode reward and summarizes the mean and standard deviation across trials.

3.2 Multi-Gap Merging Experiment

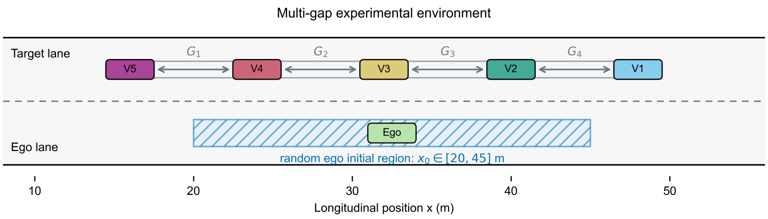

Fig. 3. multi gap environment schematic

The multi-gap environment extends the same merging task to a target lane containing five vehicles and four physical gaps. The target vehicles are initialized with a uniform base spacing, positions are $x_i(0) = 48.0 - (i-1)g_0$, $g_0 = 8.0\,\mathsf{m}$, $i = 1, \ldots, 5$.

All target vehicles start on the target-lane center with nominal speed $15.0\,\mathsf{m/s}$. The ego vehicle starts from the original lane with a larger longitudinal randomization range than in the single-gap experiment. The larger random range changes which three vehicles are nearest to the ego vehicle at the beginning of an episode. Therefore, the high-level selector must work under different local traffic configurations rather than repeatedly facing one fixed gap.

The ego initial state is $x_e(0) = 30.0 + \xi$, $\xi \sim \mathcal{U}(-10.0, 15.0)$, $y_e(0) = 2.0\,\mathsf{m}$, and $v_e(0) = 15.0\,\mathsf{m/s}$.

The four target-lane gaps are randomly adjusted over time. Every $T_g = 4.0\,\mathsf{s}$, at most two gaps are selected for modification, and each selected gap receives a desired spacing from the multiplier set $\mathcal{M} = \{0.75, 1.0, 1.25, 1.5\}$.

The leading target vehicle has zero acceleration. Each following target vehicle tracks the desired gap in front of it through a clipped proportional-derivative rule:

$$a_{i+1}(t) = \text{clip}\left(0.55\left[g_i(t) - g_{i,\text{des}}^{(k)}\right] + 1.05\dot{g}_i(t), -4.0, 4.0\right). \tag{20}$$

This mechanism produces a target lane in which gaps open, close, and reconfigure during the 40.0 s simulation. The high-level decision module selects the nearest three target-lane vehicles in longitudinal front-axle coordinate and compares the two local gaps formed by these vehicles. The multi-gap experiment is designed to evaluate generalization. The underlying strategy uses the attention strategy learned in the single-gap environment. This design tests whether a strategy that only learns $u(t)$ for a single pair of preceding and following vehicles remains effective when the high-level selector continuously provides different pairs of preceding and following vehicles. If the strategy is generalizable, then even if the gap selection and gap environment change, the ego vehicle should still be able to smoothly enter the selected gap.

The high-level ablation compares the opinion-dynamics selector with a simple maximum-score selector. The maximum-score baseline removes the high-level opinion memory and instantaneously chooses the candidate with the larger gap Confidence

This comparison was evaluated through multiple random ablation experiments. The current test setup uses $N_{\text{test}} = 100$ random runs. For each method, the total reward and the number of times the selected gap was switched were compared.

4 Results and discussion

4.1 Single-gap SAC Training

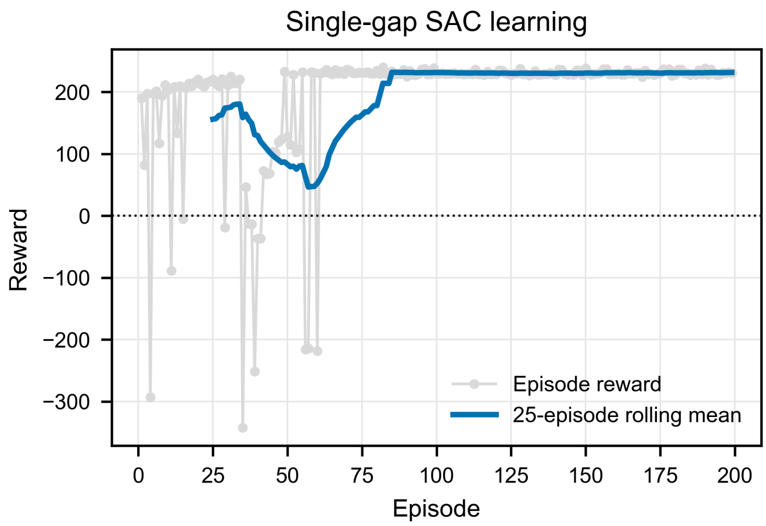

Fig. 4. single gap training reward

Training shows large reward fluctuations in the first 50 episodes because SAC explores widely. Afterward, the episodic reward stabilizes around 230 ± 10 (Fig. 4). Q1-loss decreases from 134 to below 20, and the entropy coefficient α decreases from 0.96 to 0.12. The final policy reaches progress ≥ 0.95, achieves 97.5% success, and produces no collisions.

4.2 Comparison with Baseline RBF Controller

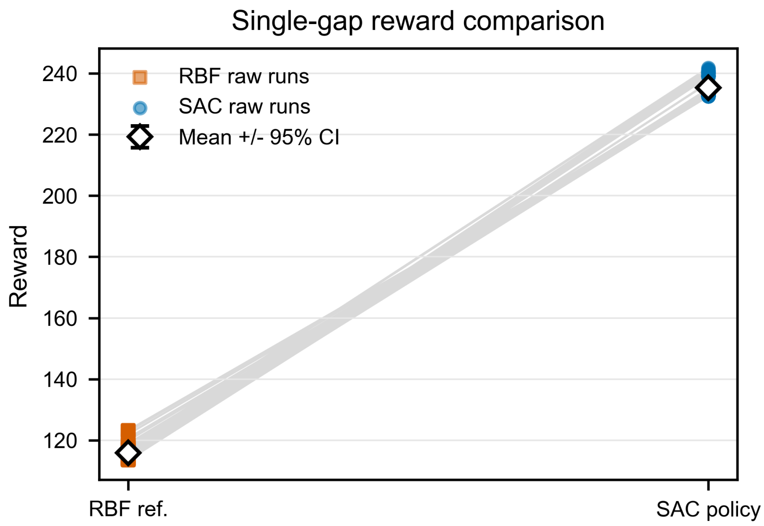

Fig. 5. Comparison between SAC and RBF reward

The SAC policy achieves a higher average episodic reward than the RBF baseline, with 233.4 versus 115.8. Both policies maintain 100% success and zero collisions, but SAC completes the task in 6.9 s, whereas RBF requires 29.45 s. This indicates that the hand-designed RBF rule is safe but conservative, while SAC learns a more efficient yet still collision-free strategy.

In terms of safety indicators, the RBF baseline maintained a larger minimum distance ($d_{\text{min}} \approx 9.7$ m), reflecting its inherent conservatism. The SAC strategy is closer to the obstacle ($d_{\text{min}} \approx 2.6$ m), but always remains outside the collision radius, proving that the learned strategy can effectively balance time efficiency and collision safety through an optimized reward structure.

Table 2. Single-gap evaluation comparison between SAC and RBF controllers

Metric	SAC Policy	RBF Baseline	Improvement
Episodic Reward ($\bar{R}$)	233.4 ± 4.2	115.8 ± 2.1	**+101.5%**
Task Time (s)	6.9 ± 0.2	29.45 ± 2.4	**-76.6%**
Success Rate	100%	100%	N/A
Collision Rate	0%	0%	N/A
Min. Distance (m)	2.62 ± 0.23	9.70 ± 0.01	N/A

4.3 Multi-Gap Generalization

The low-level SAC strategy was trained in a single-gap environment and then directly deployed to a dynamic multi-gap scenario. The experimental results show that it has good generalization ability in terms of safety and task completion: the success rate remains at a high level ($\approx$0.97). Even if the high-level selector switches, in the vast majority of cases, the low-level strategy never violates

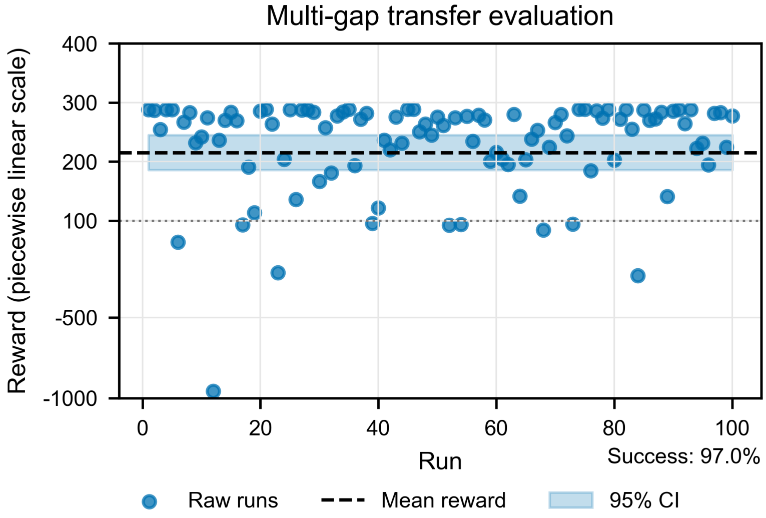

Fig. 6. The reward of the multiple gap generalization experiments

the safety boundary. These data confirm that SAC strategy generalizes beyond the training environment.

Some difficult cases still produce low reward or failure. These cases mainly arise from repeated high-level decisions that extend the manoeuvre beyond the maximum simulation time, or from overly cautious local optima that accumulate large step penalties. This suggests that the current state input is too instantaneous: it captures relative geometry and velocity but does not fully represent temporal gap switching or long-term feasibility.

4.4 High-level decision-making Ablation experiment

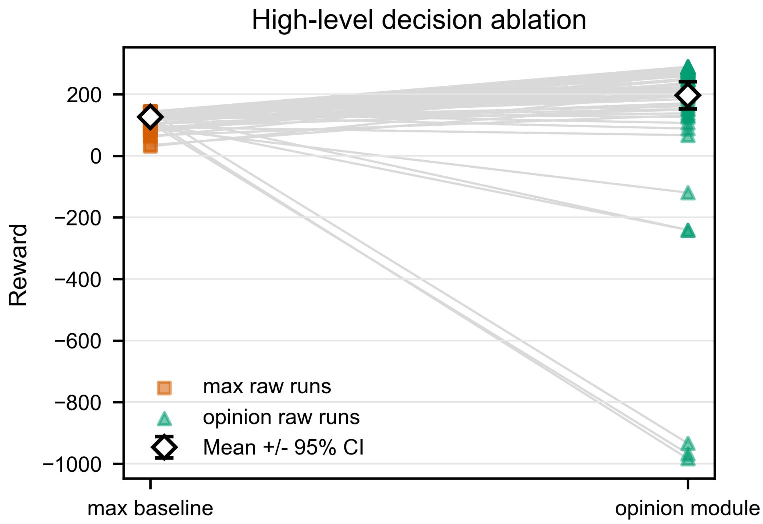

Fig. 7. Comparison of high-level strategy rewards between Opinion and Max

The opinion-dynamics selector achieves a higher mean reward than the max-score selector (+55.3%) and switches less often (0.61 versus 0.76). Although max-score sometimes finishes faster, its lower reward indicates larger safety penalties and less smooth control. The opinion-based selector therefore provides more temporally consistent guidance. And it is more stable and comfortable, despite a slightly lower success rate caused by timeout cases.

Metric	Opinion Strategy	Max Strategy	Improvement
Mean Episode Reward ($\bar{R}$)	196.8 ± 48.9	126.7 ± 28.7	**+55.3%**
Mean Switch Count	0.61 ± 0.48	0.76 ± 0.79	**-19.7%**
Mean Time (s)	12.6	8.2	N/A
Success Rate	**97%**	100%	N/A

Table 3. Multi-gap high-level decision ablation results

5 Conclusions

The two-layer opinion-dynamics framework provides an interpretable method for autonomous lane merging. The upper layer selects a local candidate gap by smoothing the confidence difference with self-updating attention, while the lower layer converts the selected gap into objective bias, learned or analytic attention, opinion-driven target interpolation, and safety-aware tracking control.

Experiments show that the SAC lower-level strategy trained in the single-gap scenario converges stably and outperforms the hand-designed RBF controller, mainly by improving efficiency while remaining collision-free. In multi-gap deployment, the same policy maintains a high success rate, showing partial generalization, but timeout failures and conservative local optima reveal limited adaptability to dynamic switching. This limitation is linked to the absence of temporal context in the state and the lower layer's inability to observe high-level switching or gap infeasibility. The ablation further shows that the opinion-based high-level selector achieves higher reward and fewer switches than the max-score baseline, confirming the value of temporally smoothed upper-layer guidance.